BIOLOGY EXAM — MARKING REPORT

SSC-I (9th Grade) — Annual Examination 2026 — Version 3

Assessment Date: 3 April 2026

Student: Seemab **Subject: Biology** **Paper: Version 3** **Total Marks: 65**

ESTIMATED TOTAL SCORE

44 / 65

68% — Grade B

Section	Score	Maximum	Percentage
A — MCQs (12 questions)	~10 (estimated*)	12	~83%
B — Short Questions (best 8 of 11)	22	24	92%
C — Long Questions (3 attempted)	12	15	80%
TOTAL	**~44**	**65**	**~68%**

*Section A (MCQs) was answered on the exam paper itself and is not included in this answer sheet. Score estimated at ~10/12.

SECTION B — Short Questions (Best 8 of 11 = 24 marks)

Q#	Topic	Score	Assessment
(i)	Biology definition + 3 divisions	**3/3**	**Excellent** Complete definition with Greek roots. Botany, Zoology, Microbiology all correct with relevant examples (photosynthesis, frog nervous system, yogurt fermentation).
(ii)	Interdisciplinary research collaboration	**3/3**	**Excellent** Good definition. Biophysics (muscles, nerve impulses, X-ray diffraction/DNA) and Biochemistry (chemical reactions in organisms, protein/nucleic acid structure) are valid textbook examples.
(iii) OR	Four kingdoms of Domain Eukarya	**2/3**	**Needs detail** All four kingdoms named correctly. Descriptions are too brief. Missing: specific examples (Amoeba, Paramecium, mushroom, yeast, moss, insects), nutrition modes for each kingdom.
(iv) OR	Prokaryotic vs Eukaryotic cells	**3/3**	**Excellent** Comprehensive comparison covering nucleus, membrane-bound organelles, size (1–10μm), DNA shape (circular vs linear), ribosomes (70S vs 80S), and examples. Well structured.
(v)	Mesophyll vs Epidermal cells	**2.5/3**	**Good** Location, chloroplasts, and function compared well. Missing: specialised epidermal cells — **guard cells** (stomata) and **root hair cells** (absorption).
(vi) OR	Define tissue + 4 animal tissue types	**3/3**	**Excellent** Clear definition. All four types (epithelial, muscle, nerve, connective) described with structure, function, types/examples. Well organised answer.
(vii)	Nucleic acids, purines & pyrimidines	**2/3**	**Error** Definition and examples correct. However, **ring structures were swapped** — she wrote purines have single ring and pyrimidines have double ring (the opposite is true). Missing base-pairing rules (A–T, G–C).
(viii)	Three types of carbohydrates	**2/3**	**Error** Content is correct but **column headers swapped** — polysaccharide definition written under disaccharide heading and vice

			versa. Formulae and examples match the (wrong) headings consistently. Missing sucrose (most important disaccharide).
(ix)	Four careers in biology	**3/3**	**Excellent** Four careers listed (Biotechnology, Fisheries, Medicine/Surgery, Horticulture). Biotechnology and Medicine described in good detail with correct information.
(x) OR	Define data, hypothesis, vector	**2.5/3**	**Good** All three definitions correct (data = facts/observations, hypothesis = tentative/testable/falsifiable, vector = transmitter organism). Missing specific examples: female *Anopheles* mosquito, *Aedes aegypti*.
(xi) OR	Binomial nomenclature	**2/3**	**Needs detail** Correct: Carolus Linnaeus, two Latin names, genus (capital) + species (lowercase). Examples (*Homo sapiens*, *Solanum tuberosum*) good. Missing: Linnaeus dates (1707–1778), **importance** (universal system, avoids confusion), italicisation rule.

Best 8 scores: 3 + 3 + 3 + 3 + 3 + 2.5 + 2.5 + 2 = **22 / 24 (92%)** — Strong performance in Section B.

SECTION C — Long Questions (3 × 5 = 15 marks)

Q.3 OR — Biological Method / Discovery of Malaria Cause — 4.5 / 5

Strong Very thorough answer covering all steps of the biological method:

✔ Recognition of problem (malaria since ancient times, cause unknown)

✔ Observations (4 key points: marshy areas, drinking water, quinine, Plasmodium in blood)
✔ Hypothesis (Laveran, 1882: "Plasmodium is the cause of malaria")
✔ Deduction ("If Plasmodium is the cause, all ill persons should have it in blood")
✔ Experiment (100 patients + 100 healthy control group)
✔ Results (almost all patients had Plasmodium, 7 healthy had it — incubation period, later became ill)
✔ Conclusion ("Results were convincing to prove the hypothesis")
✔ Reporting (results announced worldwide)

✘ **Missing:** Spread of malaria section — A.F.A. King (1883) suggested mosquitoes spread malaria, **Ronald Ross (1897)** proved female *Anopheles* mosquito transmits malaria. This is worth ~0.5 marks.

Q.4 OR — Proteins: Composition, Structure & Functions — 3.5 / 5

Good Solid answer with some gaps:

✔ Chemical composition (C, H, O, N, sometimes P, S, Fe) — correct

✔ Amino acid structure (alpha carbon, 4 groups: –H, $–NH_2$, –COOH, –R group) — detailed and correct
✔ 20–25 types of amino acids, R-group determines properties — correct
✔ Peptide bonds between amino acids — correct
✔ Four structural levels named: primary, secondary, tertiary, quaternary

✘ **Secondary structure:** Must include **alpha helix** (coiled spring shape) and **beta pleated sheet** (zigzag folding) — these are key terms examiners look for. Only wrote "folding with extra bonds".
✘ **Functions:** Listed antibodies, histones, and enzymes — valid but the expected answers were: (1) **Structural** (collagen in skin/bones, keratin in hair/nails), (2) **Enzymatic** (amylase breaks starch), (3) **Transport** (haemoglobin carries O_2). These are the top-scoring function examples.

Q.6 OR — RNA Types, Central Dogma, Transcription & Translation — 4 / 5

Strong Comprehensive answer:

✔ mRNA (carries message from DNA in nucleus to cytoplasm for protein synthesis)

✔ tRNA (transfers amino acids to ribosomes during protein synthesis)
✔ rRNA (component of ribosomes, combines with ribosomal proteins — "factory of proteins")
✔ Central dogma: DNA → RNA → Protein (flow of biological information)
✔ Transcription: DNA strands unwind, one strand (coding strand) copied into mRNA, strands rejoin
✔ Translation: mRNA moves to cytoplasm, binds to ribosome, tRNA brings amino acids as per mRNA code, polypeptide chain formed

✘ **Missing:** Gene expression = transcription + translation (she mentioned this but could elaborate). Could mention RNA polymerase enzyme and start/stop codons for full marks.

Section C total: 4.5 + 3.5 + 4 = **12 / 15 (80%)**

KEY MISTAKES TO REVIEW

1. Purines vs Pyrimidines (Q.vii) — Ring structures were swapped.
✘ **Seemab wrote:** Purines = single ring, Pyrimidines = double ring
✔ **Correct:** **Purines = DOUBLE ring** (Adenine, Guanine) = larger in size. **Pyrimidines = SINGLE ring** (Cytosine, Thymine, Uracil) = smaller in size.
*Memory trick: P**u**ri**n**es are P**U**RE and big (double ring). Pyrimidi**n**es are tiny (single ring).*

2. Carbohydrates Table (Q.viii) — Column headers accidentally swapped.
✘ **Seemab wrote:** Polysaccharide definition under "Disaccharides" column and vice versa
✔ **Remember:** **Di** = two (2 monosaccharides joined, $C_{12}H_{22}O_{11}$, e.g. sucrose, lactose, maltose). **Poly** = many (long chains, $(C_6H_{10}O_5)n$, e.g. starch, cellulose, glycogen).
*Also remember **sucrose** (glucose + fructose) = most common disaccharide (table sugar).*

3. Protein Secondary Structure (Q.4 OR) — Missing key terminology.
✔ **Must mention:** (1) **Alpha helix** = coiled spring shape, stabilised by hydrogen bonds. (2) **Beta pleated sheet** = zigzag/folded shape, also stabilised by hydrogen bonds. These are the two forms of secondary structure that examiners always look for.

4. Missing Specific Examples Throughout — Several answers lost marks due to lack of textbook examples.
✔ **Key examples to memorise:**
- Protista: *Amoeba*, *Paramecium*, algae • Fungi: mushroom, yeast, bread mold
- Vectors: female *Anopheles* mosquito (malaria), *Aedes aegypti* (dengue)
- Linnaeus: **1707–1778**, Swedish botanist • Protein functions: collagen, keratin, haemoglobin

5. Malaria Spread (Q.3 OR) — Missing the final part about how malaria spreads.
✔ **Must include:** A.F.A. King (**1883**) — observed mosquitoes might spread malaria. **Ronald Ross** (**1897**) — proved experimentally that the female *Anopheles* mosquito transmits the *Plasmodium* parasite. He first demonstrated this in birds using *Culex* mosquitoes.

OVERALL ASSESSMENT

Strengths:

✔ Strong understanding of core concepts across all chapters
✔ Section B performance is excellent (92%) — well-structured answers with good detail
✔ Q3 OR (biological method) was outstanding — covered all steps systematically
✔ Good use of tables and comparisons (prokaryotic/eukaryotic, mesophyll/epidermal)
✔ Q6 OR (RNA/central dogma) shows strong molecular biology understanding
✔ Attempted all 11 Section B questions + 3 Section C questions

Areas for Improvement:

✘ Double-check table headers and column labels before moving on
✘ Memorise specific textbook facts: dates, scientist names, examples
✘ Learn key terminology precisely (alpha helix, beta pleated sheet, base-pairing rules)
✘ Always include 2–3 specific examples for each point in long questions

Potential with corrections:

If the five mistakes above are corrected, Seemab can easily score **50–52 / 65 (77–80%)** — a solid **Grade A**.